

LPBF Processing of CP1 Aluminum Alloy

Run	VED (J/mm ³)	P (W)	v (mm/s)	h (mm)	t (mm)	Interlayer rotation	Mark
1	37.5	150	2000	0.05	0.04	67°	Point 1
2	55.0	220	2000	0.05	0.04	67°	Point 2
2							
3	68.8	275	2000	0.05	0.04	67°	Point 3
4	125.0	275	1100	0.05	0.04	67°	
5	200.0	275	688	0.05	0.04	67°	
6	300.0	275	458	0.05	0.04	67°	
7	37.5	150	2000	0.05	0.04	0°	
8	55.0	220	2000	0.05	0.04	0°	
9	68.8	275	2000	0.05	0.04	0°	
10	37.5	150	2000	0.05	0.04	67°	Remelting 75W, 2000 mm/s, 0.05mm
11	55.0	220	2000	0.05	0.04	67°	Remelting 110W, 2000 mm/s, 0.05mm
12	68.8	275	2000	0.05	0.04	67°	Remelting 137W, 2000 mm/s, 0.05mm
2	55.0	220	2000	0.05	0.04	67°	V+, h -
13	55.0	220	1429	0.07	0.04	67°	Center point
14	55.0	220	1111	0.09	0.04	67°	V-, h +
15	55.0	157	1429	0.05	0.04	67°	P-, h -
16	55.0	283	1429	0.09	0.04	67°	P+, h +
17	55.0	283	2000	0.064	0.04	67°	V+, P+, h+
18	55.0	157	1111	0.064	0.04	67°	V-, P-, h+
19	200	275	688	0.05	0.04	67°	Beam Shaping Ring beam
20	300	275	458	0.05	0.04	67°	Beam Shaping Ring beam